

# Cartesian Generalized-Internal-Coordinate Refinement of Semiexperimental Equilibrium Structures

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## Abstract

Semiexperimental equilibrium structures combine ground-state rotational constants with computed vibrational and electronic corrections, yielding structural references that are directly comparable with equilibrium quantum-chemical geometries. Existing refinement workflows demonstrated the power of nonlinear least-squares fits of moments of inertia, predicate observations, and graphical control, but still required substantial care in the construction and maintenance of the structural parameter set. Here we introduce a Cartesian generalized-internal-coordinate (GIC) formulation implemented in Merlino4. The input model is defined by a versioned self-contained job file containing keywords, Cartesian parent geometry, optional fixed-parameter definitions, and isotopologue tables with definitions, rotational constants, vibrational corrections, electronic corrections, substitutions, and uncertainties; separate reusable isotopologue files remain supported. Gaussian Cartesian input files with ModRedundant freeze records are also accepted for interoperability, while Z-matrix inputs are deliberately excluded. The fitted variables are generated automatically, reduced to a non-redundant set, and symmetrized after automatic point-group identification. Ring coordinates are built

from endocyclic torsional coordinates and the analogous valence-angle combinations, not from Cremer-Pople variables. The solver fits moments of inertia by default, supports vibrational and electronic corrections, propagates experimental uncertainties to GIC and topological parameters, includes soft quantum-mechanical predicates, and allows chemically meaningful classes of parameters to be refined together or kept fixed. A Kraitchman-coordinate diagnostic and human-readable text, HTML, CSV, and L<sup>A</sup>T<sub>E</sub>X outputs are generated for each run. The resulting workflow preserves the spectroscopic logic of molecular-structure refinement while removing user-dependent coordinate construction and making the entire procedure reproducible from both command line and graphical interfaces.

## 1 Introduction

Accurate equilibrium geometries are central to rotational spectroscopy, force-field validation, and the assessment of electronic-structure methods. Rotational spectra provide highly precise ground-state constants, but the associated structures are vibrationally averaged. The semiexperimental strategy addresses this limitation by correcting experimental constants for vibrational and, when needed, electronic effects before fitting an equilibrium geometry.

The MSR program represented an important step in this direction by combining non-linear least-squares refinement of moments of inertia, advanced error analysis, predicate observations, automatic generation of symmetry-adapted internal coordinates, and a graphical interface.<sup>1</sup> That design established several principles that remain essential: the fit should be performed in a stable observable space, missing experimental information can be complemented by weighted predicates, and uncertainty propagation must be part of the standard output rather than a post-processing option.

Merlino4 keeps these principles but changes the coordinate layer. The molecular structure is read as Cartesian coordinates, while the variables optimized in the least-squares problem are generalized internal coordinates generated automatically from topology, rings, and sym-

metry information. As in other modern coordinate generators, the primitive coordinate list is built without manual intervention; the distinctive step is the subsequent reduction to a non-redundant active set and the symmetry adaptation obtained after automatic point-group identification. This provides a common coordinate layer for Gaussian input generation, Wilson GF analysis, VPT2/VCI workflows, ring-puckering paths, and semiexperimental refinement.

The present article describes the new semiexperimental structure solver and its integration into the broader GIC infrastructure. Particular attention is paid to four extensions motivated by practical spectroscopy: grouped refinement of parameter classes such as CH stretches, blocking of poorly determined hydrogen-angle classes, explicit comparison with Kraitchman substitution coordinates, and complete graphical access to the same options available from the command line.

## 2 Theory and Method

### 2.1 Semiexperimental Observables

For each isotopologue  $s$ , the experimental ground-state rotational constants are converted to equilibrium targets by subtracting computed corrections,

$$B_{\alpha,e}^{(s)} = B_{\alpha,0}^{(s)} - \Delta B_{\alpha,\text{vib}}^{(s)} - \Delta B_{\alpha,\text{elec}}^{(s)}, \quad \alpha \in \{a, b, c\}. \quad (1)$$

Merlino fits principal moments of inertia by default,

$$I_{\alpha}^{(s)} = \frac{K}{B_{\alpha,e}^{(s)}}, \quad (2)$$

where  $K$  is the standard rotational conversion constant. Fitting moments avoids the reciprocal amplification that can occur when optimizing directly in rotational-constant space. Direct fits of rotational constants remain available for diagnostic comparisons, with auto-

matic selection of the best-conditioned constant pair for planar molecules.

## 2.2 Automatic Non-Redundant and Symmetry-Adapted GIC Parameterization

Starting from the parent Cartesian geometry, Merlino constructs a molecular topology and a primitive internal-coordinate list automatically. This first construction step is deliberately similar in spirit to other black-box internal-coordinate generators: bonds, valence angles, linear bends, torsions, out-of-plane terms, and ring-local primitives are generated from the molecular graph. Merlino then performs two additional operations before the coordinates enter the least-squares model. First, the primitive set is reduced to a non-redundant GIC transformation,

$$\mathbf{q} = \mathbf{U}^T \mathbf{s}, \tag{3}$$

where  $\mathbf{s}$  is the vector of primitive coordinates. Second, the point group is identified from the Cartesian structure and used to symmetrize the non-redundant GICs without mixing different coordinate types. Stretches, bends, torsions, ring-valence combinations, and other coordinate families are therefore reduced and symmetrized within their own homogeneous blocks.

For rings, Merlino does not use Cremer-Pople coordinates in the GIC layer. Ring puckering and ring deformation are represented by combinations of endocyclic dihedral angles and by analogous combinations of endocyclic valence angles. This choice keeps the variables in the same generalized-internal-coordinate algebra as the rest of the molecule and avoids switching to a separate ring-specific coordinate convention during structure refinement.

The same GIC machinery is used by the Merlino GF and Gaussian-input workflows, ensuring that structural refinement and vibrational analysis operate on consistent coordinates.

The Cartesian displacement associated with a GIC correction is obtained from the ana-

lytic Wilson matrix,

$$\Delta \mathbf{x} = \mathbf{B}_q^+ \Delta \mathbf{q}, \quad \mathbf{B}_q = \mathbf{U}^T \mathbf{B}_s, \quad (4)$$

with the pseudoinverse evaluated under rank control. Trial steps are accepted only if they lower the weighted least-squares objective.

## 2.3 Least-Squares Model

The residual vector contains selected moments or rotational constants for all isotopologues and optional predicate observations,

$$\mathbf{r} = \mathbf{y}_{\text{obs}} - \mathbf{y}_{\text{calc}}(\mathbf{q}). \quad (5)$$

The weighted Levenberg-Marquardt step is

$$(\mathbf{J}^T \mathbf{W} \mathbf{J} + \lambda \mathbf{I}) \Delta \mathbf{p} = \mathbf{J}^T \mathbf{W} \mathbf{r}. \quad (6)$$

Experimental uncertainties in MHz are propagated to moments when the moment target is used. The final covariance is obtained from the inverse weighted normal matrix scaled by the reduced chi-square, and the Gauss-Newton Hessian is diagonalized to classify the stationary point.

## 2.4 Predicate Observations, Constraints, and Parameter Classes

Quantum-mechanical estimates of individual GIC parameters can be introduced as soft predicate observations with user-defined uncertainties. This follows the MSR philosophy while keeping all predicates in the generated GIC label space.

Merlino4 additionally supports parameter classes. A class can be refined as a single shared variable,

$$\Delta q_i = \Delta p_c \quad i \in c, \quad (7)$$

or held fixed. This is useful when deuterated isotopologues are unavailable for all hydrogen sites. Typical examples are grouped CH stretches and blocked XYH or XH<sub>2</sub> angle families. The Jacobian is compressed by a class transformation matrix before solving the normal equations, and uncertainties are expanded back to all members of the class in the reported parameter table.

Hard constraints are kept separate from predicates. Parameters can be frozen by matching explicit GIC-label patterns, and Gaussian-style ModRedundant freeze records can be read from Cartesian input files or from the Merlino job file. The recognized records are bond, angle, dihedral, out-of-plane, and linear-bending freezes. They are converted to GIC-label patterns before the active totally symmetric subspace is selected, so constrained coordinates remain visible in the output but are excluded from the normal equations.

## 2.5 Kraitchman Diagnostic

For isotopologues containing a single substitution, Merlino computes substitution coordinates from the Kraitchman equations and compares their absolute values with the fitted coordinates in the parent principal-axis frame. This diagnostic is intentionally separate from the fit: Kraitchman coordinates provide valuable assignment checks, but they do not retain coordinate signs and do not exploit the full correlated least-squares model.

## 3 Implementation

The solver is implemented in Python and mirrored by independent Fortran77 numerical kernels for analytic **B** rows, rotational constants, weighted normal equations, and parameter-class compression. The standard input uses two files. The first is a versioned Merlino semiexperimental job file, identified by the schema string `merlino.semiexp.job.v1`. It contains fit keywords, the ordered Cartesian parent geometry in Å, fixed GIC-label patterns, optional ModRedundant freeze cards, predicate observations, and parameter-class definitions. The

second file contains isotopologue observations in TOML, JSON, or CSV form. For each isotopologue it stores ground-state rotational constants, isotope substitutions, vibrational corrections, electronic corrections, correction conventions, and optional experimental uncertainties.

Gaussian Cartesian `.com/.gjf` files remain accepted as geometry inputs when the isotopologue data are supplied separately. In that mode the Cartesian block supplies atom numbering and coordinates, and any ModRedundant `F` cards following the coordinate block define frozen parameters. Gaussian `Z`-matrices, dummy atoms, and MSR-style variable blocks are rejected because they reintroduce the coordinate-template ambiguity that the Cartesian/GIC formulation is designed to remove.

The graphical dashboard exposes the same controls as the command-line interface: job-file generation, parent Cartesian file selection, isotopologue table editing, backend request, observable, rotational-component policy, fixed parameters, predicate observations, and parameter classes. Jobs launched from the GUI are asynchronous and write a manifest with checksums and numerical diagnostics.

In addition to CSV files, each run writes a plain-text report, an HTML report, and  $\text{\LaTeX}$  table fragments. The text report is intended for rapid inspection and records convergence diagnostics, corrected experimental and calculated rotational constants, final topological bond lengths, angles, and dihedrals with propagated errors, fitted GIC parameters, and residuals. The CSV files remain the machine-readable source for downstream processing.

## 4 Differences from MSR

The new implementation is directly inspired by the semiexperimental strategy and predicate-observation formalism used in MSR,<sup>1</sup> but differs in several operational and methodological aspects.

Table 1: Conceptual comparison between the MSR workflow and the Merlino4 GIC workflow.

| Feature                | MSR                           | Merlino4                                                                       |
|------------------------|-------------------------------|--------------------------------------------------------------------------------|
| Structural input       | user molecular model          | Cartesian parent geometry                                                      |
| File contract          | program-specific input        | versioned self-contained job file; reusable isotopologue files optional        |
| Fit coordinates        | internal/symmetry coordinates | automatic GICs, reduced to non-redundant symmetry-adapted blocks               |
| Observable target      | moments of inertia            | moments by default; constants optional                                         |
| Predicate observations | yes                           | yes, in GIC label space                                                        |
| Fixed parameters       | manual coordinate selection   | GIC-label patterns or ModRedundant freeze cards                                |
| Parameter classes      | limited/manual                | shared or fixed GIC classes                                                    |
| Kraitchman check       | external/manual               | generated diagnostic CSV                                                       |
| GUI control            | yes                           | same options as CLI, asynchronous jobs, job-file export                        |
| Human-readable output  | program reports               | text, HTML, CSV, manifest, and $\LaTeX$ tables                                 |
| Ring coordinates       | internal coordinate dependent | endocyclic dihedral and valence-angle combinations, not Cremer-Pople variables |
| Cross-workflow reuse   | structure refinement          | GIC, GF/PED, VPT2/VCI, DVR interfaces                                          |

## 5 Planned Validation

The first validation set will include the molecules used in the MSR article where the required rotational data and vibrational corrections are available, followed by ring and fused-ring cases that stress automatic GIC generation, non-redundant pruning, symmetry adaptation, and ring coordinates built from endocyclic torsions and valence angles. For each molecule we will provide the versioned job file, the reusable isotopologue observation file, residuals in moments and rotational constants, fitted GIC parameters with propagated uncertainties, final topological parameters with errors, correlation matrices, stationary-point diagnostics, and Kraitchman comparisons for all single-substitution isotopologues.



## 6 Conclusions

Merlino4 provides a semiexperimental refinement workflow in which versioned input files, Cartesian geometry, automatic GIC construction, non-redundant reduction, automatic symmetry adaptation, analytic **B** matrices, robust least squares, predicate observations, class constraints, Kraitchman diagnostics, text/HTML/CSV/L<sup>A</sup>T<sub>E</sub>X reports, and GUI execution are part of one reproducible pipeline. The approach keeps the spectroscopic strengths of MSR while extending the coordinate model to the same GIC infrastructure used by the rest of Merlino.

## References

- (1) Mendolicchio, M.; Penocchio, E.; Licari, D.; Tasinato, N.; Barone, V. Development and Implementation of Advanced Fitting Methods for the Calculation of Accurate Molecular Structures. *Journal of Chemical Theory and Computation* **2017**, *13*, 3060–3075.